Accident Analysis and Prevention 161 (2021) 106350

Contents lists available at ScienceDirect

Accident Analysis and Prevention

journal homepage: www.elsevier.com/locate/aap

A relative approach to the validation of surrogate measures of safety

Carl Johnsson *, Aliaksei Laureshyn, Carmelo Dágostino

Department of Technology and Society, Faculty of Engineering, LTH Lund University, Lund, Sweden

ARTICLE INFO

Keywords:
Surrogate measures of safety
Bicyclists
Relative validation

ABSTRACT

Surrogate measures of safety (SMoS) are meant to be an alternative/complement to crash data that enables a proactive approach to traffic safety. Validity is a fundamental property of SMoS that describes how well they reflect the quality of interest – traffic safety. However, typical validation studies are extremely resource demanding. Therefore, the aim of this study was to explore a relative approach to validity that uses fewer resources and does not rely on crash records from the observed locations. The core idea of a relative approach to the validation of SMoS is that while SMoS might not accurately estimate the expected number of crashes, they still work well for comparisons between different sites or between conditions at the same site. Based on this concept, we propose a method in which a ground truth based on literature can be used as a baseline that can then be compared to the safety analysis of an SMoS study. Using the proposed method, a case study of six intersections in northern Europe was conducted, focusing on bicycle infrastructure. The results indicate that the minimum time-to-collision indicator is in line with the ground truth, whereas the post-encroachment time indicator is not.

1. Introduction

Surrogate measures of safety (SMoS) are meant to be an alternative/complement to crash data that enables a proactive approach to traffic safety. The main idea behind SMoS is that near-crash traffic events can be used as surrogates for real crashes. By studying these events, it is possible to learn about the safety of a specific traffic system. The advantage of using SMoS is that near crashes occur much more frequently than crashes, which makes it easier (1) to directly observe these events in traffic and (2) to conduct safety studies over relatively short periods.

The underlying assumption is that there is continuity between traffic events along a certain severity dimension (Fig. 1). Some events are 'closer to a crash' than others, and there is a relationship between the severity of the events and the frequency at which they can be observed in traffic.

Validity is a fundamental property of SMoS that describes how well they reflect the quality of interest – traffic safety. Validity is often interpreted as the ability of SMoS to estimate the expected number of crashes at a given site. Validation studies embracing this definition focus on finding relationships between the frequency of near crashes and the number of actual or expected crashes estimated using other methods (e.g. using crash history data from a given site or constructing a crash prediction function using crash data from many comparable sites).

While this validity interpretation is theoretically the most correct and the most valuable from the perspective of the future use of SMoS, it is a very complex problem to solve. There are several pitfalls and challenges when designing classical validation studies, including low crash counts in general, uneven under-reporting, the necessity of proper control of the exposure, and subjectivity in 'near crash' selection, which might be involved in direct or indirect ways. These considerations render any attempt at validation extremely resource-demanding, thus making validation studies quite rare. In a recent literature review, Johnsson, Laureshyn, et al. (2018) provided an overview of previous validation studies that actually attempted to relate SMoS and crash data, most of which were conducted several decades ago. The authors concluded that there is still a lack of validation studies for many indicators and that most previous studies relied mostly on human observers and not on computer vision techniques that are typically used now (Laureshyn et al., 2016). There is a definite need for more validation studies, but there is also a need for a more practical approach to conducting validation studies that can reduce their 'cost'.

1.1. Aims

The primary aim of this study was to explore and test a relative approach to the validation of SMoS. This approach is less strict but possibly more practical, since it does not require the use of any crash

* Corresponding author.
E-mail addresses: carl.johnsson@tft.lth.se (C. Johnsson), aliaksei.laureshyn@tft.lth.se (A. Laureshyn), carmelo.dagostino@tft.lth.se (C. Dágostino).

https://doi.org/10.1016/j.aap.2021.106350
Received 20 August 2020; Received in revised form 12 July 2021; Accepted 8 August 2021
Available online 21 August 2021
0001-4575/© 2021 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

records, thus eliminating many concerns typically encountered in SMoS validation studies. This approach was tested on two of the most commonly used indicators in SMoS studies: post-encroachment time (PET) and minimum time-to-collision (MTTC). The secondary aim was to explore the relationship between elementary units of exposure (such as meetings and interactions) and SMoS and examine how this relationship should be considered when studying the validity of SMoS.

1.2. The notion of relative validity

The core idea of a relative approach to the validity of SMoS is that, while certain SMoS might not estimate the expected number of crashes, they still work well for comparisons between different sites or between conditions at the same site (De Ceunynck, 2017). Basically, relatively valid SMoS can reflect the direction of a change in safety (improvement or deterioration) but cannot describe its magnitude. Despite this limitation, such information can in many cases still be of high value. Some examples are quick feedback on the introduction of a new safety measure with no documented effects or safety assessments in conditions where no crash data can be obtained (either because safety is already high and thus crash records are limited or because a crash reporting system is not functioning properly).

The use of relative validity tackles some of the practical issues related to the validation of SMoS. To relate 'near crashes' to crash estimates, a large number of data points (e.g. locations and conditions) are usually required due to the nature of the statistical tools used. For example, using variations of the ratio of observed near misses to registered crash counts as a measure of validity, as described by Hauer and Gårder (1986), necessitates meaningful variation calculations for a large number of sites. In a relative validation study, the comparison of only two sites can contribute a useful piece of knowledge. As such small-scale studies are easier to conduct, it can be expected that knowledge can be accumulated faster, albeit in smaller steps.

Another implication is the position of the threshold between near misses and 'normal events'. Theoretically, if the goal is to estimate the expected number of crashes, the strictest possible threshold is to be preferred, as these events should be the 'closest to a crash'. This makes the frequency of relevant events low and the necessary observation period long. For a relatively valid SMoS, the threshold should be just high enough to distinguish between the safety levels of two sites/conditions and will presumably be significantly more inclusive. In practice, that would mean that such validation studies – and practical safety assessments based on SMoS – could be conducted over shorter periods. Thus, more such studies can be expected.

The notions of relative validity and safety rankings are obviously not new concepts and are commonly used in traffic safety research. However, a literature review by Laureshyn et al. (2016) found no previous validation studies using a relative approach to the validity of SMoS, even though it can offer a less resource-intensive alternative to the traditional methods, as well as a theoretical framework for determining suitable threshold values and identifying the 'best' of several indicators using the frequency of observations.

1.3. Elementary units of exposure and SMOS

'Near crashes' are not a measure of exposure (Güttinger, 1984; Hauer, 1982). The purpose of exposure is to 'take account of the amount of opportunity for accidents' (Chapman, 1973), while 'near-crashes', like actual crashes, are the result of both the exposure and the crash risk at a given site. While it is quite common to define exposure in terms of numbers of vehicles or kilometres travelled, it makes more sense to use a definition in which a unit of exposure is actually an event that can be seen as an opportunity for a crash to occur (Elvik, 2015). A meeting, that is, the simultaneous presence of two road users heading towards a common conflict area, is a good candidate.

Both crashes and 'near crashes' are subsets of a larger set that includes all events with any probability of a crash (i.e. exposure). Fig. 2 illustrates this relationship between elementary units of exposure, crashes, and 'near crashes', at different threshold levels (assuming that the SMoS used to define 'near crashes' accurately describes the safety hierarchy discussed in the previous section).

The event-based definition of exposure provides a clear and logical link between exposure, risk (probability of an exposure unit developing into a crash), and crashes. It also allows for controlling for exposure in SMoS validation studies in a transparent way. To explain this, let us consider a validation study in which a certain definition of a 'near crash' is found to correlate with crashes. The question should be whether this correlation is a result of functional similarities and possibly causal relationships or simply a result of the fact that both are subsets of elementary events constituting the exposure. As illustrated in Fig. 2, if the threshold used is low, it will include a large number of elementary traffic events and will thus not significantly differ from the exposure. Yet, it might appear as a well-functioning definition of a 'near accident' just because there is a close relationship between crashes and exposure.

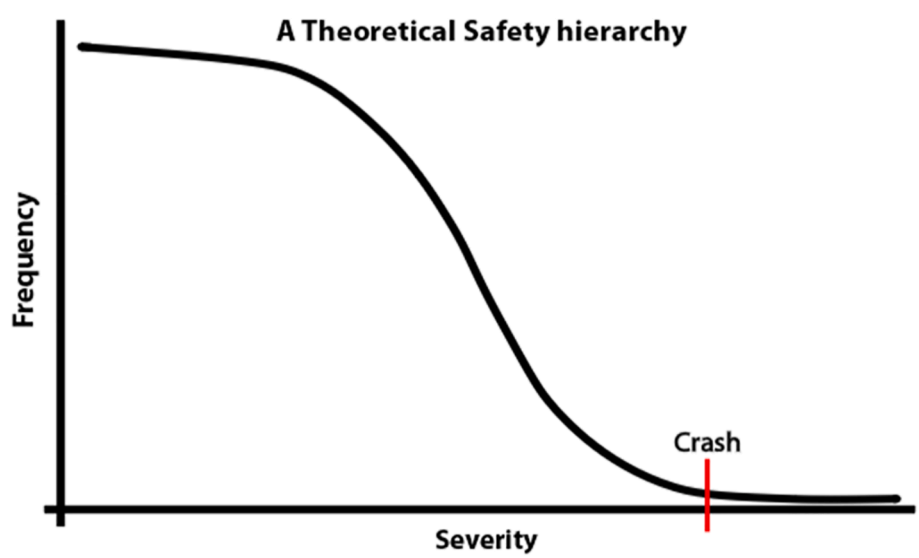

Fig. 1. A theoretical safety hierarchy showing the relationship between the frequency of events and their severity.

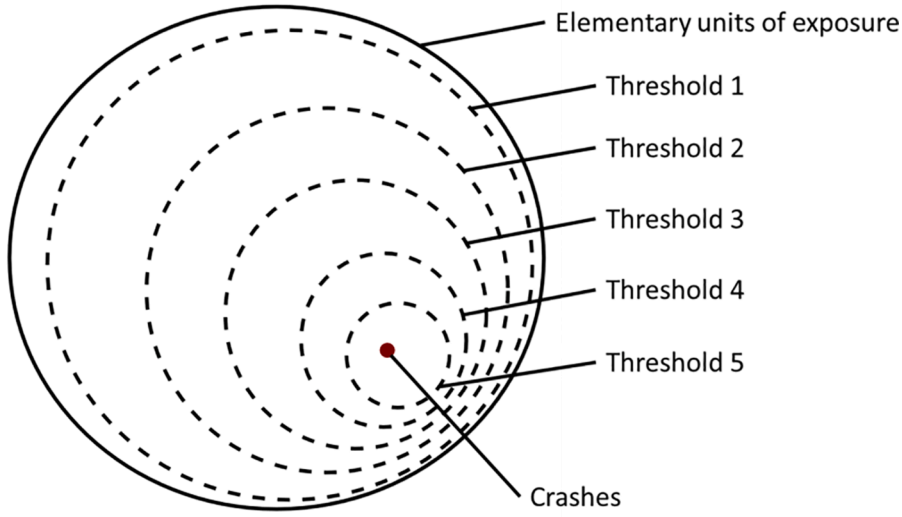

Fig. 2. The theoretical relationship between elementary units of exposure, SMoS at different threshold values, and crashes. Adopted from Amundsen and Hyden (1977, p. 137).

2. A method for testing the relative validity of SMoS

Using the concept of relative validity and the idea of elementary units of exposure, it is possible to construct a practical approach to testing the validity of SMoS. Such an approach would consist of two main parts, the ground truth and the SMoS study, and would follow the process shown in Fig. 3.

Ground truth refers to a known safety ranking between two different types of infrastructure designs. The main aim of the ground truth is to establish a 'correct answer' that can be used to test the SMoS. This known safety ranking needs to be established by previous studies using other methods. There are two main approaches to establishing a ground truth. One approach is to rely on the general knowledge gained from numerous previous studies that support a general safety ranking of different designs. The other approach is to instead rely on a single previous study estimating the safety ranking at specific locations and use those specific locations to test the SMoS.

Once the ground truth has been established, an SMoS study can be conducted. The study aims to test whether a certain indicator (using different threshold values) can be used to distinguish the safety ranking established by the ground truth. The selection of the locations to observe in the study depends on how the ground truth was established. If the ground truth consists of a general safety ranking that differs between two designs, the locations should be chosen to represent the typical design as accurately as possible, considering as many confounding variables as possible. However, if the ground truth was established by a single study, the SMoS study should be conducted at the specific locations investigated by that previous study.

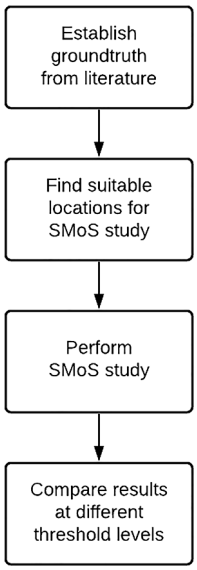

Fig. 3. The proposed approach to validity.

To properly study the effect of a specific indicator (using a specific threshold), the SMoS study should use an event-based definition of exposure when counting the number of opportunities for a crash. Furthermore, it should test several threshold values and can include multiple indicators. If several indicators/thresholds produce the correct safety rankings, the most preferable one is the indicator/threshold with the highest observed frequency.

3. A case study using MTTC and PET at bicycle crossings

This section provides an example of a validation study of the MTTC and PET indicators using the proposed relative approach to validity. These indicators were selected because they are the most commonly used indicators (Laureshyn et al., 2016) and can be calculated based on cyclist movement.

3.1. Safety ranking – ground truth

The study focused on interactions between right-turning motor vehicles and cyclists at intersections with separated cycle crossings or cycle lanes (Fig. 4). Specifically, it examined three intersections with separated cycle crossings located in the Netherlands and three intersections with cycle lanes located in Denmark. The included locations are a subset of observed locations from seven countries studied in the European project In-depth Understanding of Accident Causation for Vulnerable Road Users (InDeV, 2015–2018). The Netherlands and Denmark were chosen because of their high numbers of cyclists and the consistent design of their respective intersections.

Separated cycle crossings are characterised by a recessed cycle track in conjunction with a pedestrian crossing in which the cyclists are physically separated from motor vehicles before and after the intersection. In contrast, cycle lanes are separated from motor vehicles only with painted lines that continue before and after the intersection. According

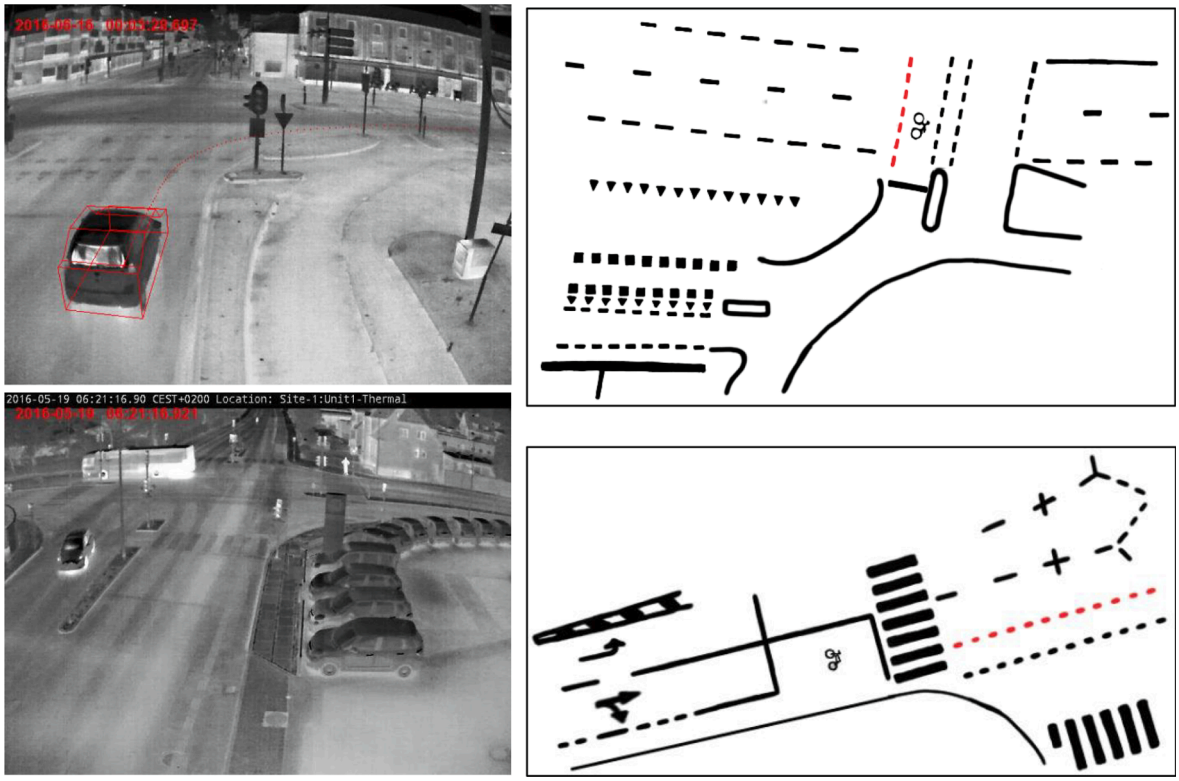

Fig. 4. The two studied designs: separated cycle tracks (top) and cycle lane (bottom). The images on the left are camera views of two intersections. The images on the right show schemas of the designs. The red dotted lines show where the speed measurements were performed. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to a systematic literature review by Prati et al. (2018), there is evidence that separated cycle crossings are less safe than cycle lanes. This seems to hold true when focusing solely on right-turning motor vehicles (Jensen & Transportation Research Board, 2008). A reason for this difference in safety might be that drivers turning right tend to pay less attention to cyclists coming from the right (Summala et al., 1996). Despite the evidence suggesting that cycle lanes are generally safer than separated cycle crossings, Prati et al. (2018) found that many other factors can affect cyclists' safety. Moreover, Mulvaney et al. (2015) noted a general lack of high-quality evidence to draw definite conclusions regarding the role of cycling infrastructure in cycling collisions.

Speed seems to be an exception to this general lack of high-quality evidence. Several reviews have found that reducing the speed of motor vehicles has a positive impact on safety (Aarts & van Schagen, 2006; Mulvaney et al., 2015; Prati et al., 2018; Thomas & DeRobertis, 2013). Fig. 5 shows the speed of unhindered motor vehicles making a right turn at the studied intersections, measured when the motor vehicles first started to cross the cycle paths (indicated by the red lines in Fig. 4). The speed was measured from video recordings using trajectories and gates created in the T-Analyst software (Johnsson, Norén, et al., 2018). The speed distribution shows that the crossing speeds at the intersections in the Netherlands, with separated cycle crossings, are

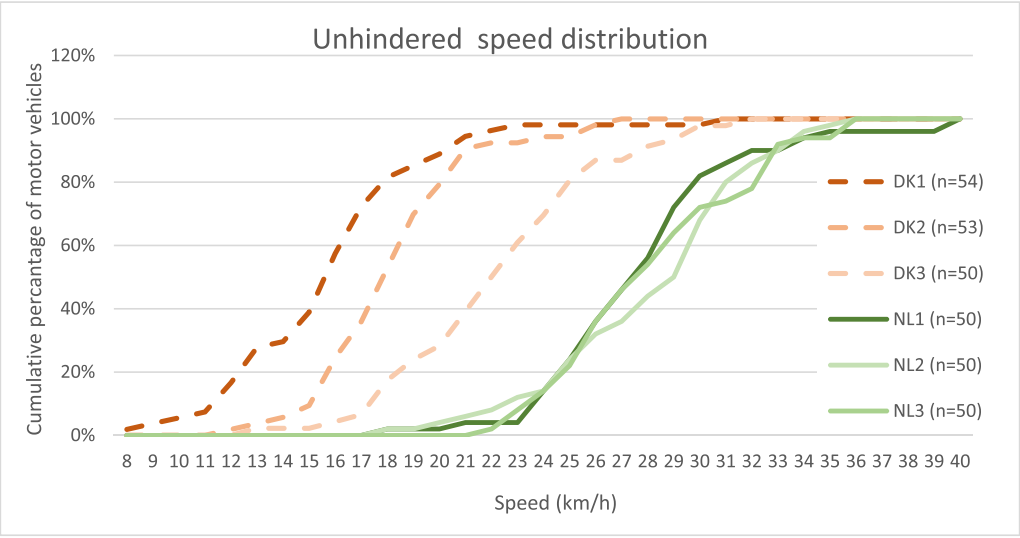

Fig. 5. Cumulative speeds of unhindered motor vehicles at the observed locations, measured when the motor vehicles started to cross the cycle paths.

significantly higher than those at the intersections in Denmark. In conclusion, both the designs and the speed measurements indicate that the three studied intersections in the Netherlands are less safe than the three Danish intersections.

Nevertheless, some caveats regarding the selected traffic systems should be noted. While the intersections in the Netherlands are remarkably similar, the Danish intersections exhibit some discrepancies. The main difference is that the first intersection (DK1) is smaller than the others, with only one lane for motor vehicles, whereas all the others (including the ones in the Netherlands) have two. Some smaller differences were also observed. The first Danish intersection (DK1) did not have blue paint on the cycle lane, and the second intersection (DK2) had a bike box, unlike the other two.

3.2. The SMoS study

Based on video recordings at the six intersections included in the study, manual identification of all encounters between right-turning motor vehicles and bicycles over a 24-hour period was performed. The trajectory of each encounter was then manually created using the T-Analyst software (Johnsson, Norén, et al., 2018) and was then used to calculate its PET and MTTC values. The following sections provide a brief description of each indicator and explain how encounters were defined. The results are then presented.

3.3. Minimum time-to-collision

Time-to-collision (TTC) is the time remaining to an impact, assuming that the road users continue to travel as initially planned (Hayward, 1971). The TTC can be calculated provided that a collision course is present. The MTTC is the lowest value obtained during a meeting. This also means that an MTTC value can be calculated only if the road users were on a collision course at some point during the meeting. Typical thresholds range between 1.5 and 3 s (Laureshyn et al., 2016).

An important practical issue related to the use of the TTC is that it requires a procedure for calculating the future motion of a road user. In this study, the calculations were made based on the assumption that the road user would travel along the revealed trajectory and maintain a constant speed. The detailed procedure was described by Laureshyn et al. (2010).

3.4. Post-encroachment time

Post-encroachment time, introduced by Allen et al. (1978), is a measure of 'near misses', that is, situations in which a collision is avoided by a relatively small margin. It is calculated as the time from the moment the first road user leaves the path of the second user until the moment the second road user reaches the path of the first user. In other words, it indicates the margin by which they miss each other. Using a threshold value, it is possible to obtain a definition of safety–critical events using the PET, in which smaller values are considered more severe. Typical thresholds range from 1 to 3 s (Laureshyn et al., 2016).

3.5. Encounters

The main exposure measure used in this study was encounters (or meetings) between bicyclists and motor vehicles. The basic idea is to include only situations in which a right-turning motor vehicle and a crossing bicyclist are simultaneously present at the intersection. Each encounter included only one motor vehicle and one vulnerable road user. If there were multiple vulnerable road users passing in front of the motor vehicle, only the one closest to the motor vehicle while it was still in motion was selected. Similarly, if there were several queuing motor vehicles, only the frontmost was considered.

3.6. Results

Table 1 shows the numbers of encounters during the 24-hour observational period and the corresponding numbers of safety–critical events. Overall, more encounters occurred at the Dutch intersections, with the first intersection having considerably more events than the others. Regarding the exposure levels at the different locations, Table 2 shows the number of critical events per encounter at the various threshold levels. Fig. 6 and Fig. 7 show graphical representations of the results displayed in Table 2.

An examination of the risk estimates shown in Fig. 6 and Fig. 7 reveals that the two indicators do not agree. Using MTTC, the SMoS analysis indicated that the intersections in the Netherlands were less safe, which is consistent with the previously established ground truth. Considering only the mean values, the result was the same regardless of the threshold value used. However, when looking at the individual sites, only the threshold values of 4 and 3 s showed that the intersections in the Netherlands are consistently less safe than those in Denmark. This is confirmed by a comparison between the intersections in the two countries using Welch's *t*-test, which generated significant p-values (less than 0.05) only for MTTC with thresholds of 3 s, 4 s, and ∞. This result is somewhat unexpected, since SMoS theory holds that lower thresholds should provide a more accurate safety analysis. A possible explanation for this is a significant decrease in the number of events using the threshold values of 2 and 1.5 s. It is possible that the low number of events caused a wide random variation, making the results somewhat unclear.

On the other hand, the SMoS analysis using PET produced different results. First, there seems to be a very strong correlation between encounters and critical events with threshold values above 2 s. This might indicate that at least with these high threshold values, the PET indicator does not actually measure safety but mostly exposure. Second, the results are not in line with the ground truth. Using threshold values of 2 and 1 s, the Danish sites were found to be consistently less safe than those in the Netherlands (p less than 0.05).

4. Discussion

This paper argues that a relative approach to validity is a compromise that can allow less resource intensive SMoS validity tests by relying on established safety rankings from previous research. The proposed approach can be applied to test SMoS relatively quickly without the need for resource-intensive work focusing on establishing the expected crash frequency.

However, the proposed approach has three limitations. The first limitation is that any validity test based on previous studies relies on the validity and reliability of those studies. Therefore, the validity of the SMoS relies on the strength of the argument for the ground truth. This case study, for example, relied on several literature reviews summarising the existing knowledge of the relationship between safety and bicycle infrastructure design. Although the fact that these literature reviews agree on the role of speed indicates that the proposed ground truth makes sense, some doubt regarding the infrastructure design itself remains.

The second limitation of the proposed approach is that there is no real yardstick for measuring how many locations are suitable for a validity test. It is obviously preferable for an SMoS study to include many locations, although the proposed approach also works with fewer locations. This problem might be alleviated by an increased number of validation studies; and since the main advantage of the relative approach is that it allows less resource-intensive validity studies, it can hopefully result in more studies. A large number of smaller validity tests will also allow indicators to be tested in various scenarios, including different designs and road user types. This is important if an indicator is to be universal (i.e. usable in all kinds of traffic situations), which is harder to test with few large studies.

Table 1
Numbers of encounters and safety–critical events at the six intersections over 24 h according to different threshold values.

Site	Encounters	MTTC < ∞	MTTC ≤ 4 s	MTTC ≤ 3 s	MTTC ≤ 2 s	MTTC ≤ 1.5 s	PET ≤ 5 s	PET ≤ 4 s	PET ≤ 3 s	PET ≤ 2 s	PET ≤ 1 s
Denmark 1	205	28	23	18	7	3	195	193	189	157	54
Denmark 2	179	63	45	23	5	3	179	179	176	165	64
Denmark 3	205	17	10	7	3	2	195	195	195	179	54
Denmark total	589	108	78	48	15	8	569	567	560	501	172
Netherlands 1	317	219	185	112	29	6	292	284	272	223	62
Netherlands 2	190	90	84	58	20	11	188	185	177	137	27
Netherlands 3	155	56	50	34	4	3	151	150	141	111	30
Netherlands total	662	365	319	204	53	20	631	619	590	471	119

MTTC < ∞ is the number of events in which a collision course was found during an encounter, regardless of the actual TTC value. MTTC: minimum time-to-collision; PET: post-encroachment time.

Table 2
Numbers of critical events per encounter at the six intersections over 24 h according to different threshold values.

Site	MTTC < ∞	MTTC ≤ 4 s	MTTC ≤ 3 s	MTTC ≤ 2 s	MTTC ≤ 1.5 s	PET ≤ 5 s	PET ≤ 4 s	PET ≤ 3 s	PET ≤ 2 s	PET ≤ 1 s
Denmark 1	0.14	0.11	0.09	0.03	0.01	0.95	0.94	0.92	0.77	0.26
Denmark 2	0.35	0.25	0.13	0.03	0.02	1.00	1.00	0.98	0.92	0.36
Denmark 3	0.08	0.05	0.03	0.01	0.01	0.95	0.95	0.95	0.87	0.26
Denmark mean (SD)	0.19 (0.14)	0.14 (0.10)	0.08 (0.05)	0.03 (0.01)	0.01 (0.00)	0.97 (0.03)	0.96 (0.03)	0.95 (0.03)	0.85 (0.08)	0.29 (0.05)
Netherlands 1	0.69	0.58	0.35	0.09	0.02	0.92	0.90	0.86	0.70	0.20
Netherlands 2	0.47	0.44	0.31	0.11	0.06	0.99	0.97	0.93	0.72	0.14
Netherlands 3	0.36	0.32	0.22	0.03	0.02	0.97	0.97	0.91	0.72	0.19
Netherlands mean (SD)	0.51 (0.17)	0.45 (0.13)	0.29 (0.07)	0.07 (0.04)	0.03 (0.02)	0.96 (0.04)	0.95 (0.04)	0.90 (0.04)	0.71 (0.01)	0.18 (0.03)
P-value*	**0.03**	**0.02**	**0.01**	**0.07**	**0.13**	**0.41**	**0.30**	**0.07**	**0.04**	**0.03**

MTTC < ∞ is the number of events in which a collision course was found during an encounter, regardless of the actual TTC value.
* Obtained from comparisons between the Danish and Dutch intersections using Welch's *t*-test.
MTTC: minimum time-to-collision; PET: post-encroachment time; SD: standard deviation.

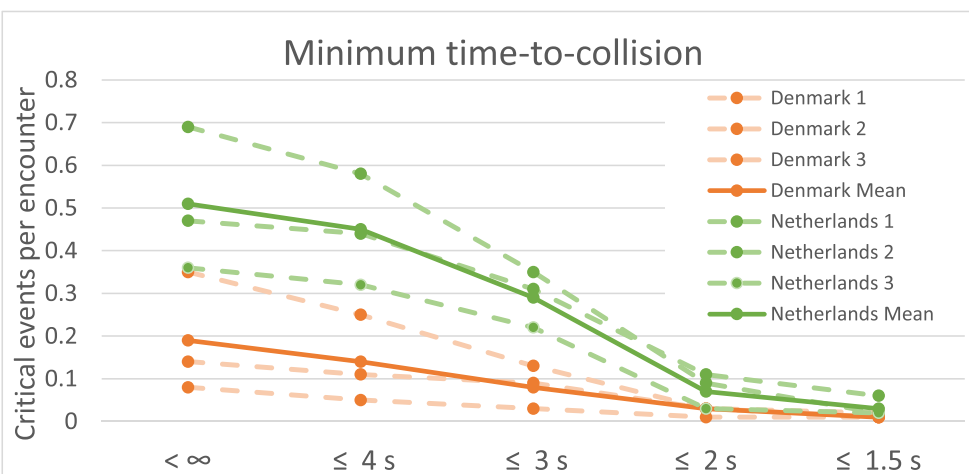

Fig. 6. Numbers of critical events per encounter at the six intersections using minimum time-to-collision.

The third limitation is that by focusing solely on the safety ranking, differences in the magnitude of critical events between two locations or designs are not considered. Such differences are typically and intuitively used to discuss differences in safety between different locations or designs. For example, this case study found that there are on average 3.2 times more critical events per encounter at the intersections in the Netherlands compared to the Danish intersections (using MTTC with a threshold of 4 s). However, this result would be disregarded, and only the fact that Denmark has more safety–critical events would be considered. To overcome this limitation, the concept of an SMoS study could be divided into two types: the classical SMoS study and the relative SMoS study. The classical approach would focus on estimating the frequency of safety–critical events with low threshold values according to the idea of safety hierarchy (i.e. lower threshold values are 'closer' to crashes), while the relative approach would focus on the highest possible threshold that still allows an accurate safety ranking analysis. This distinction makes theoretical sense in that the optimal threshold would be different for the two types and practical sense in that it would enable a shorter and more easily applicable relative approach and a longer and more resource-intensive classical approach to SMoS.

Finally, while this research focuses on validation of SMoS without the use of crash data, that does not mean that crash data could not be used to better support the proposed method in future research. For example, actual number of crashes at the different sites might be used to establish a safety ranking between the sites which could further strengthen the case for a relative approach to validity. However, this also causes some of the concerns regarding the gathering of crash data to re-emerge. This was not attempted in this this study since the crash data

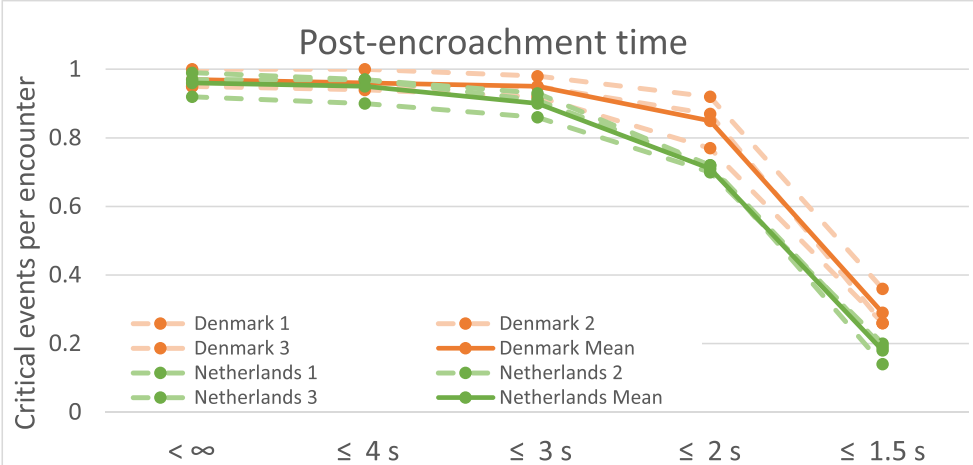

Fig. 7. Numbers of critical events per encounter at the six intersections using post-encroachment time.

from the Netherlands did not contain information about the specific location of the crash nor what manoeuvre the road users were performing when they collided.

5. Conclusion

The main conclusions of this study can be summarised as follows:

1. The relative approach to validity offers a less resource-intensive method for studying the validity of SMoS, accumulating knowledge with small, but hopefully more frequent, steps.
2. Using the relative approach to validity, the optimal threshold value would become less strict compared to the optimal threshold according to the classical approach. This suggests that dividing future SMoS studies into a shorter and more easily applicable relative version and a longer and more resource-intensive classical version might be useful.
3. Controlling for event-based exposure measures, such as meetings between road users, is important due to their strong correlation with safety–critical events.
4. The results of this study indicate that the MTTC indicator with a threshold value between 3 and 4 s is consistent with the ground truth, suggesting that it correctly measures traffic safety.
5. The results also show that the PET indicator fails to agree with the ground truth at any threshold value. The results further indicate a very strong correlation between exposure and PET at threshold values above 2 s.

CRediT authorship contribution statement

Carl Johnsson: Conceptualization, Methodology, Software, Investigation, Resources, Data curation, Writing – review & editing. **Aliaksei Laureshyn:** Conceptualization, Methodology, Software, Resources, Supervision, Project administration, Funding acquisition. **Carmelo Dágostino:** Methodology, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work received funding from the European Union's Horizon 2020 research and innovation programme (grant agreement number 635895).

References

Aarts, L., van Schagen, I., 2006. Driving speed and the risk of road crashes: A review. Accid. Anal. Prev. 38 (2), 215–224. https://doi.org/10.1016/j.aap.2005.07.004.
Allen, B.L., Shin, B.T., Cooper, P.J., 1978. Analysis of traffic conflicts and collisions. Transp. Res. Rec. 667, 67–74.
Amundsen, F.H., Hyden, C., 1977. Proceedings: First workshop on traffic conflicts. First workshop on traffic conflicts, Oslo, Norway.
Chapman, R., 1973. The concept of exposure. Accid. Anal. Prev. 5 (2), 95–110. https://doi.org/10.1016/0001-4575(73)90018-3.
De Ceunynck, T. (2017). Defining and applying surrogate safety measures and behavioural indicators through site-based observations [Doctoral dissertation, Lund University]. Lund University Research Portal. https://portal.research.lu.se/portal/files/30184385/170823Dissertation_TimDeCeunynck_final_inclcover.pdf.
Elvik, R., 2015. Some implications of an event-based definition of exposure to the risk of road accident. Accid. Anal. Prev. 76, 15–24. https://doi.org/10.1016/j.aap.2014.12.011.
Güttinger, V.A. (1984). Conflict observation in theory and in practice. In E. Asmussen (Ed.), International calibration Study of Traffic Conflict Techniques. NATO ASI Series (Series F: Computer and Systems Sciences), vol 5. Springer. 10.1007/978-3-642-82109-7_3.
Hauer, E., 1982. Traffic conflicts and exposure. Accid. Anal. Prev. 14 (5), 359–364. https://doi.org/10.1016/0001-4575(82)90014-8.
Hauer, E., Gårder, P., 1986. Research into the validity of the traffic conflict technique. Accid. Anal. Prev. 18 (6), 471–481. https://doi.org/10.1016/0001-4575(86)90020-5.
Hayward, J.C., 1971. *Near misses as a measure of safety at urban intersections* [Unpublished master's thesis]. Pennsylvania State University.
Jensen, S.U., & Transportation Research Board (2008). Bicycle tracks and lanes: A before-and-after study.
Johnsson, C., Laureshyn, A., De Ceunynck, T., 2018a. In search of surrogate safety indicators for vulnerable road users: A review of surrogate safety indicators. Transport Reviews 38 (6), 765–785. https://doi.org/10.1080/01441647.2018.1442888.
Johnsson, C., Norén, H., Laureshyn, A., & Ivina, D. (2018). T-Analyst - semi-automated tool for traffic conflict anlysis (InDeV, Horizon 2020 project. Deliverable 6.1.).
Laureshyn, A., Johnsson, C., De Ceunynck, T., Svensson, Å., de Goede, M., Saunier, N., Włodarek, P., van der Horst, A. R. A., & Daniels, S. (2016). Review of current study methods for VRU safety. Appendix 6 – Scoping review: Surrogate measures of safety in site-based road traffic observations (Deliverable 2.1 – part 4).
Laureshyn, A., Svensson, Åse, Hydén, C., 2010. Evaluation of traffic safety, based on micro-level behavioural data: Theoretical framework and first implementation. Accid. Anal. Prev. 42 (6), 1637–1646. https://doi.org/10.1016/j.aap.2010.03.021.
Mulvaney, C. A., Smith, S., Watson, M. C., Parkin, J., Coupland, C., Miller, P., Kendrick, D., & McClintock, H. (2015). Cycling infrastructure for reducing cycling injuries in cyclists. Cochrane Database of Systematic Reviews (12). 10.1002/14651858.CD010415.pub2.
Prati, G., Marín Puchades, V., De Angelis, M., Fraboni, F., Pietrantoni, L., 2018. Factors contributing to bicycle–motorised vehicle collisions: A systematic literature review. Transport Reviews 38 (2), 184–208. https://doi.org/10.1080/01441647.2017.1314391.
Summala, H., Pasanen, E., Räsänen, M., Sievänen, J., 1996. Bicycle accidents and drivers' visual search at left and right turns. Accid. Anal. Prev. 28 (2), 147–153.
Thomas, B., DeRobertis, M., 2013. The safety of urban cycle tracks: A review of the literature. Accid. Anal. Prev. 52, 219–227. https://doi.org/10.1016/j.aap.2012.12.017.